High rotation in the plane: the absorptivity explains the asymmetry of up to 100 to 1 between the senses

Willian Wistuba Melo da Cunha
Self-taught physics enthusiast

We take the test of the three-body flux theory with the measured absorptivity to the high-rotation regime studied by Dandekar et al., with the mass sets 10-15-20 and 5-15-25 and the rotation of their initial condition (L/L0 of 1.06 and 1.03). Without absorptivity, the theory predicts equal numbers of prograde and retrograde final pairs, and they measured ratios of 2.08 and 144. In free falls with rotation, we measure the absorptivity separately for the two senses of rotation of the pair and test the prediction in 8 preregistered tests, at 95% confidence and with Holm correction. All pass: the absorptivity predicts, with no free parameter, ratios of about 2 to 1 in 10-15-20 and of 79 to 105 to 1 in 5-15-25, as well as the escape fractions of each body. Against the published numbers, the prediction passes in 10-15-20, where the predicted light-body fraction, 0.782, coincides with the published one, 0.783, and it explains as an effect of the absorptivity the agreement with the three-dimensional prediction noted by the authors. In 5-15-25, the published numbers lie above the prediction; this difference has the scale of the sensitivity to the chaos selection criterion, which we measure, and it vanishes when their initial condition is analyzed with our criterion (exploratory).

Keywords: three-body problem; flux theory; chaotic absorptivity; angular momentum; prograde and retrograde orbits; planar problem.

ORCID: 0009-0008-2875-6607

Contact: profwistuba01@gmail.com

Published: 6 November 2026. Version 1. Not peer reviewed.

Series: The three-body problem, part 17 of 17. https://fisica-revisitada.vercel.app/en/articles/assimetria-com-rotacao-alta/

Translated from the original in Brazilian Portuguese.

I. INTRODUCTION

Dandekar et al. tested Kol's flux theory in the planar three-body problem with 1 million simulations per mass set, always with total angular momentum above L_0 [1, 4]. Without measuring the absorptivity, the planar theory got two things wrong. The escape fractions came closer to the three-dimensional prediction, within 1%, than to the planar one, something the authors called surprising. And the ratio between prograde and retrograde final pairs, which the theory fixes at 1, ranged from 0.90 to 144, growing with the contrast between the masses. As causes, they pointed to the emissivity-blind approximation and to a possible incomplete mixing of phase space in the plane.

Note FR-TC-16 showed that, at low rotation, the absorptivity measured separately for the two senses of rotation of the pair predicts the prograde fraction [5]. This note takes the same method to the regime of Dandekar et al., with two of their mass sets and the rotation of the initial condition they used. The question, fixed before the data, is whether the measured absorptivity explains the two discrepancies.

II. METHOD

Theory. In the plane, the flux weight for a pair with relative energy $x = E_B/E$, eccentricity e, and sense s relative to the total rotation is $(L_0^2/2)\,x^{-2}\,e/\sqrt{1-e^2}$, equal in the two senses and independent of L [4, 5]. The prediction for each outcome is the sum, over the three escape channels and the two senses, of $(m_a m_b)^3/(m_a+m_b)$ times this weight times the measured absorptivity $A_k(x,e,s;L)$. Without absorptivity, the fraction of each body equals the L_0^2 of the remaining pair divided by the sum of the three, and the prograde fraction equals 1/2.

Sets and rotation. We use the sets 10-15-20 and 5-15-25 of Dandekar et al., the latter being the one with the highest contrast and the greatest asymmetry [4]. The rotation is that of their initial condition: L/L_0 = 1.0567 and 1.0263, with $L_0 = \sqrt{k_B/(-2E)}$ of the heaviest pair (Table 1 of Dandekar et al.).

- **Free falls with rotation:** 12,000 per set, with the positions of the free falls from rest and a rigid rotation that gives exactly the L/L_0 above [5].
- **Absorptivity:** the three channels of each set in the two senses, 132 cells of x and e and 77 cells of nearly radial pairs, 200 runs each (250,800 per set). No cell was left unreached.
- **Chaos:** at least k scrambles before the first excursion, in absorption, and in the last episode before the escape, in emission, with k = 3 and 4 [6].

The integrator is the chain integrator with algorithmic regularization of the previous notes [7]. Of the free falls, 1.7% and 1.9% did not end by the time limit; the relative energy error stayed below 2×10^{-8}. Of the 501,600 absorptivity runs, 7 had a high energy error, and none counted as chaos.

Preregistered tests. The first family compares the prediction with our free falls: prograde fraction (exact test, with the binomial drawn from the distribution of the prediction obtained by 500 resamplings of the absorptivity) and escape fractions (χ^2 with 2 degrees of freedom), for the two sets and the two values of k. The second family compares the prediction with the

Table I. Our free falls with the rotation of Dandekar et al., measured against the prediction with the measured absorptivity. Prograde fraction, ratio between prograde and retrograde pairs, and escape fraction of the lightest body. With the Holm correction over the 8 tests, all pass. Without absorptivity, the theory predicts a prograde fraction of 0.5 and a light-body fraction of 0.658 (10-15-20) and 0.939 (5-15-25).

Set (n)	k	Prograde	Ratio	Light body
10-15-20 (1,172)	3	0.675 / 0.692	2.1 / 2.2	0.777 / 0.782
10-15-20 (511)	4	0.685 / 0.654	2.2 / 1.9	0.783 / 0.770
5-15-25 (1,832)	3	0.987 / 0.989	79 / 86	0.939 / 0.954
5-15-25 (849)	4	0.991 / 0.988	105 / 82	0.940 / 0.950

numbers published by Dandekar et al.: the prograde fraction and the escape fraction of the lightest body. Each family has 8 tests, with Holm correction at 95% [8]. It was fixed beforehand that, if the theory holds, the chaotic outcome depends only on the masses, the energy, and the angular momentum, and both families should pass; a failure only in the second would point to a difference in the selection criterion or in the ensemble.

Exploratory. We also ran 12,000 runs per set in the initial condition of Dandekar et al. (circular pair with semi-major axis 5, lightest body at rest at 100 from the center of mass of the pair, random phase), with our integrator and our chaos yardstick. Since the L/L_0 of this condition is exactly the one used in the absorptivity, the same prediction applies. Finally, we applied to these runs the time-based chaos cut they used (lifetime of at least 80 years and at least 10 years between the end of the last excursion and the escape).

III. RESULTS

A. The absorptivity explains the asymmetry and the escapes in our simulations

The first family passes all 8 tests (Table I). In 5-15-25 the asymmetry is extreme, with 79 to 105 prograde pairs per retrograde one, and the prediction with the measured absorptivity gives 82 to 86, with no free parameter. In 10-15-20 the measured ratio is 2.1 to 2.2 against 1.9 to 2.2 predicted. The escape fractions also match; the smallest p, 0.008 in 5-15-25 with $k = 3$, lies above the threshold of 0.00625 of the correction. The shape of the final eccentricity passes in all four cases (exploratory; KS from 0.029 to 0.051).

B. Comparison with the published numbers

The second family passes in 10-15-20 and fails in 5-15-25 (Table II). In 10-15-20, the predicted light-body fraction, 0.782, coincides with the published one, 0.7827, and the predicted prograde fraction, 0.692, is compatible with the published one, 0.675 ($p = 0.14$). The "surprising" agreement with the three-dimensional prediction (0.7834) is, in this set, an effect of the absorptivity. In 5-15-25, the published numbers (0.975 for the light body and a ratio of 144) lie above the prediction (0.954 and 86), with $p < 0.001$.

Table II. Numbers published by Dandekar et al., prediction with the measured absorptivity, and our runs in their initial condition, all with $k = 3$ in the last two columns. By the Holm correction, the prediction passes in 10-15-20 and fails in 5-15-25 against the published numbers.

Set	Quantity	Published	Predicted	Their condition
10-15-20	prograde	0.675	0.692	0.684
10-15-20	light body	0.783	0.782	0.767
5-15-25	prograde	0.993	0.989	0.990
5-15-25	light body	0.975	0.954	0.962

C. The difference in 5-15-25 depends on the selection criterion

In the initial condition of Dandekar et al., with our chaos yardstick, the prediction matches again in 5-15-25: prograde fraction 0.990 and 0.985 (k = 3 and 4) against 0.989 and 0.988 predicted, and light body 0.962 and 0.949 against 0.954 and 0.950 (Fig. 1). In 10-15-20, in this condition, the light body lies 1.1 and 2.0 standard deviations below the prediction with k = 3 and 4.

The time-based chaos cut they used cannot be reproduced exactly from what is published, because the paper does not define how the end of an excursion is marked nor which instant counts as the lifetime. Applied to our runs with two reasonable readings of the lifetime instant (the last close encounter or the detection of the escape), the cut retains 2.0% or 8.6% of the systems in 5-15-25 (they report 6%), and the light-body fraction ranges from 0.934 to 0.981. This range, obtained only by changing the selection criterion, is larger than the difference between our prediction and the published number. The group's earlier code, attached to the thesis of Mama Zard, uses another initial condition and applies no chaos cut [9].

IV. DISCUSSION

In the regime of Dandekar et al., the absorptivity measured per sense is enough to predict, in our simulations, the asymmetry between the senses up to about 100 to 1 and the escape fractions, in both sets and with both chaos criteria. This supports the first cause they suggested, the emissivity-blind approximation, and shows no sign of incomplete mixing in the measured observables.

The remaining difference, in 5-15-25 and against the published numbers, has the scale of the sensitivity to the selection criterion that we measured, and it disappears when their initial condition is analyzed with our yardstick. With what is published, it is not possible to decide whether it comes only from the time-based cut. One possible explanation is that, in 5-15-25, part of the light-body escapes occur after short chaotic episodes, which a time-based cut may accept and a scramble-based cut rejects; this note does not test this hypothesis.

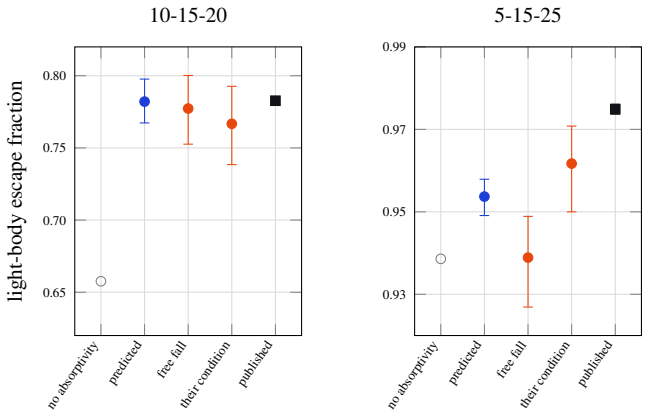


Fig. 1. Fraction of escapes of the lightest body with the rotation L/L_0 of Dandekar et al., in the two sets. From left to right: prediction without absorptivity, prediction with the measured absorptivity, our free falls with rotation, our runs in their initial condition, and the published value. Measurements with at least three scrambles in the last episode; 95% bars (Wilson for the measurements, resampling for the prediction, published standard error).

Limitations:

- two of the eight mass sets of Dandekar et al. and a single rotation level per set;
- the comparison with the published numbers depends on selection criteria that are not the same and are not fully described;
- with four scrambles, 442 to 849 escapes remain per case;
- the physical mechanism of the dependence of the absorptivity on the sense was not analyzed.

V. CONCLUSIONS

- With the rotation and the sets 10-15-20 and 5-15-25 of Dandekar et al., flux theory with the measured absorptivity passes the 8 preregistered tests against our simulations: prograde fraction and escape fractions, at 95% confidence and with Holm correction.
- The asymmetry between prograde and retrograde pairs, from 2 to 1 up to about 100 to 1, is predicted by the measured absorptivity, with no free parameter.
- In 10-15-20, the prediction reproduces the published numbers; the agreement with the three-dimensional prediction is an effect of the absorptivity.
- In 5-15-25, the published numbers lie above the prediction, but this difference has the scale of the sensitivity to the selection criterion and vanishes in their initial condition with our chaos yardstick.

TRANSPARENCY AND USE OF ARTIFICIAL INTELLIGENCE

The author is a lawyer and not a physicist. This work results from independent research, conducted with the assistance of artificial intelligence (a language model) in the literature searches, the programming of the simulations, the analysis of the results, the writing, and the translation. The author formulated the questions, followed the experiments, and is responsible for the published content.

DATA AND CODE AVAILABILITY

All numbers were obtained with scripts published at https://fisica-revisitada.vercel.app/en/code/, which run on Node.js without external dependencies and allow the results to be reproduced.

CONFLICT OF INTEREST

The author declares no conflicts of interest. This work received no funding. This text has not undergone peer review and should be read as a working note. Corrections are recorded in the note itself, with the date.

CITE AS

W. W. M. da Cunha, "High rotation in the plane: the absorptivity explains the asymmetry of up to 100 to 1 between the senses," Física Revisitada, Technical Note FR-TC-17 (2026), https://fisica-revisitada.vercel.app/en/articles/assimetria-com-rotacao-alta/.

[1] B. Kol, Flux-based statistical prediction of three-body outcomes, Celest. Mech. Dyn. Astron. 133, 17 (2021). doi.org/10.1007/s10569-021-10015-x

[2] V. Manwadkar, B. Kol, A. A. Trani, and N. W. C. Leigh, Testing the flux-based statistical prediction of the three-body problem, Mon. Not. R. Astron. Soc. 506, 692 (2021). arxiv.org/abs/2101.03661

[3] V. Manwadkar, A. A. Trani, and B. Kol, Measurement of three-body chaotic absorptivity predicts chaotic outcome distribution, Celest. Mech. Dyn. Astron. 136, 4 (2024). doi.org/10.1007/s10569-023-10174-z

[4] Y. Dandekar, E. Springer, S. Zard, A. A. Trani, B. Kol, and O. Birnholtz, Planar three-body problem: theoretical predictions and simulation results, arXiv:2608.30414 (2026). arxiv.org/abs/

2608.30414

[5] W. W. M. da Cunha, Física Revisitada, Technical Note FR-TC-16 (2026). fisica-revisitada.vercel.app/artigos/rotacao-e-sentido/

[6] W. W. M. da Cunha, Física Revisitada, Technical Note FR-TC-15 (2026). fisica-revisitada.vercel.app/artigos/onde-a-teoria-falha/

[7] S. Mikkola and K. Tanikawa, Algorithmic regularization of the few-body problem, Mon. Not. R. Astron. Soc. 310, 745 (1999). doi.org/10.1046/j.1365-8711.1999.02982.x

[8] S. Holm, A simple sequentially rejective multiple test procedure, Scand. J. Stat. 6, 65 (1979). www.jstor.org/stable/4615733

[9] S. Mama (Zard), Three-body system in two dimensions, MSc thesis, Bar-Ilan University (2024). www.biuastrograv.com/members